

MIST Student IT Portal

Product Requirement Document (PRD) — Internship Learning Management & Tracking Platform

Author: Samuel
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Target Organization: Ministry of Innovation, Science and Technology

1. Executive Summary

The **MIST Student IT Portal** is a specialized Learning Management System (LMS) designed specifically for Student Industrial Work Experience Scheme (SIWES) and IT interns accepted into the Lagos State Ministry of Innovation, Science and Technology (MIST).

The portal streamlines the management of intern curricula, structures learning paths based on department/unit allocation (e.g., Software Development, Networking, IT Support), organizes learning materials, tracks student progress in real-time, and provides administrative oversight including invitation management and student access control (the "blocking" mechanism for non-performance or security compliance).

2. Objective & Problem Statement

- **The Problem:** Managing numerous IT interns across multiple units within Alausa Secretariats makes it challenging for supervisors to track daily training compliance, standardized curricula, and overall learning progression.
- **The Solution:** A centralized web portal where MIST administrators upload official learning paths and resources, and interns systematically study and record completed modules.

3. Target User Personas

User Persona	Description & Access Level	Key Needs
MIST IT Administrator / Supervisor	Full platform access. Creates curricula, sends registration invites, blocks unfocused or inactive students, and reviews progression logs.	Simple administrative controls, instant progress overview, quick system access control switches.
SIWES / IT Student (Intern)	Authorized registered user. Can access unit-specific learning modules, download/view training materials, and mark milestones as completed.	Clear structure of expectation, easily accessible materials, progress indicators.

4. Core Technical Architecture & Schemas

The system will be powered by a lightweight, scalable **Node.js, Express, and MongoDB** backend alongside a responsive **React and Vite** frontend.

4.1 Database Design (Mongoose Schema Reference)

To ensure robust query times and cleanly maintain records, the following schemas are enforced:

```
// User Schema featuring access state representation
const userSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true },
  password: { type: String },
  role: { type: String, enum: ['admin', 'student'], default: 'student' },
  unit: { type: String, required: true },
  isBlocked: { type: Boolean, default: false },
  inviteToken: { type: String, default: null },
  tokenExpires: { type: Date, default: null },
}, { timestamps: true });
```

```
// Curriculum Schema holding hierarchical learning materials
const curriculumSchema = new mongoose.Schema({
  unit: { type: String, required: true, unique: true },
  modules: [{
    title: { type: String, required: true },
    description: { type: String },
    materials: [{
      title: { type: String, required: true },
      type: { type: String, enum: ['pdf', 'video', 'link', 'doc'], required: true },
      url: { type: String, required: true }
    }]
  }]
}, { timestamps: true });
```

5. Feature Requirements

5.1 Admin Invitation Workflow

To prevent unauthorized registrations, self-signup is disabled. Students can only register if explicitly invited by a MIST Administrator.

- **System Action:** The Admin triggers an invite via the Admin UI, sending a crypto-generated token URL (valid for 48 hours) to the student's email address.
- **Student Verification:** Clicking the URL validates the token against the database, opening a registration wizard allowing the student to set their password.

5.2 Progress Tracking & Milestones

Interns must mark off individual curriculum items as they finish reviewing them. Supervisors can instantly view overall progress in a percentage format (e.g., *65% complete*) inside their administrator dashboard tracking panel.

5.3 The Suspension/Blocking Mechanism

If an intern displays persistent lack of focus, fails to submit logs, or breaks Alausa workspace guidelines, administrators can click a **"Block Access"** button.

Important Implementation Detail: The backend authorization middleware validates the status of ``isBlocked`` on every incoming API call. If flagged true, the server returns a 403 `Forbidden` status code, and the frontend redirects the client to an account suspension advisory page immediately.

6. Non-Functional Requirements

- **Security:** Secure storage of credentials utilizing bcrypt hashing. Stateless Session Verification via JSON Web Tokens (JWT).
- **Reliability:** Offline-friendly progressive data saving (local caching) to accommodate intermittent Alausa network connections.
- **Responsive Design:** Fully responsive interface running fluidly across mobile devices, tablets, and desktop workstations.